

Geometry

Unit 12

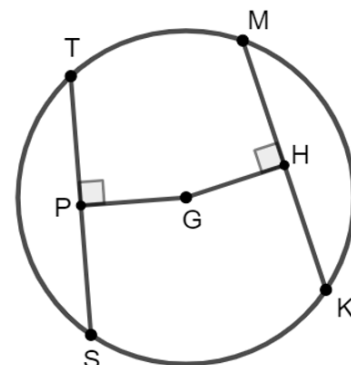
Lesson 1 Homework

Name _____

Date _____

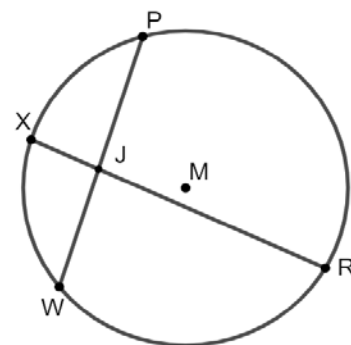
Period _____

Circle G with points T, M, K , and S on the circle are shown with chords $\overline{TS}$ and $\overline{MK}$ equidistant from the center. $TS = (12x - 5)$, $MK = (8x + 13)$, and $m\angle TPG = m\angle MHG = 90^\circ$



1. Find the value of x .
2. What is the length of chord $\overline{TS}$?

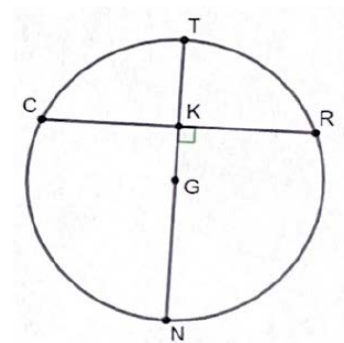
In Circle M , chords $\overline{RX}$ and $\overline{PW}$ intersect at point J . Points R, W, X , and P lie on Circle M . $RJ = 4y$, $JX = 3$, $PJ = (2y - 2)$, and $JW = 8$.



3. Solve for the value of y .
4. What is the length of $\overline{PJ}$?

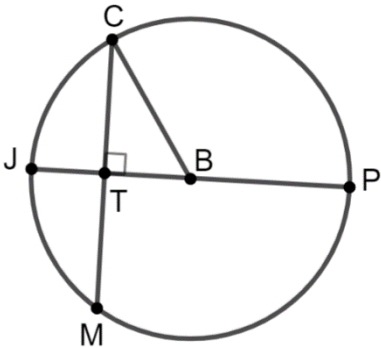
5. Find the length of chord $\overline{PW}$.

Circle G is shown where points C, T, R , and N are on the circle. The diameter, $\overline{NT}$, is a perpendicular bisector of chord $\overline{CR}$. $CR = (13x + 5)$ and $KR = (25x - 1)$.



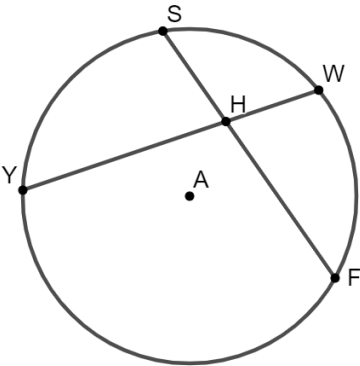
6. Calculate the value of x .
7. What is the length of chord $\overline{CR}$?

Chord $\overline{CM}$ is drawn in Circle B so that it intersects the diameter $\overline{JP}$.



8. Given that $CM = 30$ and $TB = 8$, what is the length of $\overline{BP}$?

Circle A is shown where points Y, S, W , and F lie on the circle. Chords $\overline{SF}$ and $\overline{YW}$ intersect at point H . $YH = 8$, $SH = 6$, $HW = (5x - 3)$, and $HF = (2x + 10)$. Aniyah and Colton are solving for x and their work is shown below.

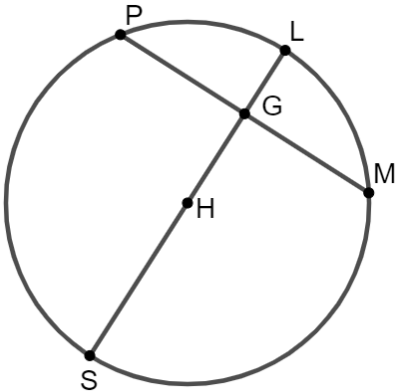


Aniyah's Work	Colton's Work
$8(5x - 3) = 6(2x + 10)$ $40x - 24 = 12x + 60$ $28x = 84$ $x = 3$	$6(5x - 3) = 8(2x + 10)$ $30x - 18 = 16x + 80$ $14x = 98$ $x = 7$

9. Whose work do you agree with? Justify your answer.

In Circle H , points P, L, M , and S lie on the circle. Chord $\overline{PM}$ intersects the diameter $\overline{SL}$ at point G and $m\angle HGP = 90^\circ$.

10. If $PM = 24$ and $HS = 14$, what is the length of $\overline{GH}$? Round to the nearest hundredth if necessary.



Geometry

Unit 12

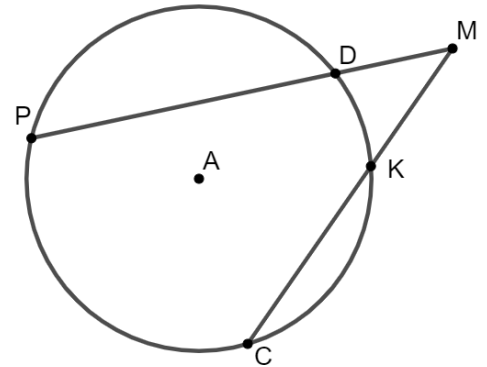
Lesson 2 Homework

Name _____

Date _____

Period _____

Circle A with secants $\overline{PM}$ and $\overline{CM}$ is shown. $\overline{PM}$ and $\overline{CM}$ intersect at point M outside the circle so that $PD = x$, $DM = 3$, $CK = 8$, and $KM = 4$.

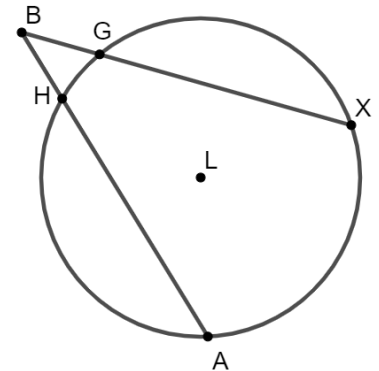


1. What is the length of secant $\overline{CM}$?

2. Find the value of x .

3. What is the length of secant $\overline{PM}$?

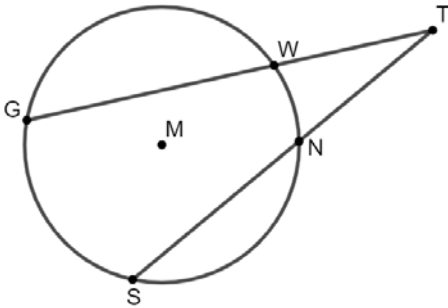
Circle L is shown, where points G , X , A , and H lie on circle L . Secants $\overline{BX}$ and $\overline{BA}$ intersect at point B outside the circle such that $BG = 6$, $GX = 20$, $BH = 4$, and $HA = (6x - 7)$.



4. What is the value of x ?

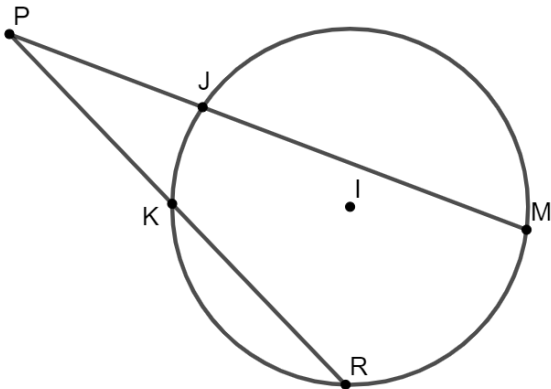
5. Calculate the length of secant $\overline{BA}$.

Circle M with secants $\overline{GT}$ and $\overline{ST}$ is shown. Secants $\overline{GT}$ and $\overline{ST}$ intersect outside the circle at point T so that $GW = 12x$, $WT = 4x$, $SN = 20$, and $NT = 16$.



6. Find the value of x .
7. What is the length of secant $\overline{GT}$?

Carter and Isabella were solving for x in circle I where points J , M , R , and K are on the circle and secants $\overline{PM}$ and $\overline{PR}$ intersect at point P outside the circle. They know $PJ = 3$, $JM = (4x - 1)$, $PK = 4$, and $KR = 2x$.

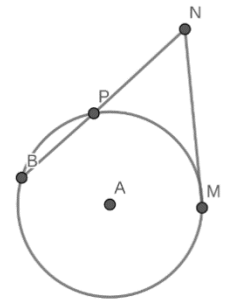


Each of their equations are set up to solve for x .

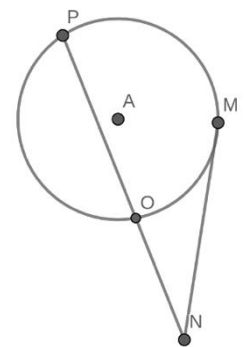
Carter's Work	Isabella's Work
$3(4x - 1) = 4(2x)$	$3(4x + 2) = 4(2x + 4)$

8. Whose work is the correct equation needed to solve for x ?
9. Use this equation to find the value of x .
10. What is the length of secant $\overline{PR}$?

1. Circle A is shown with points M , B and P on the circle. Secant $\overline{BN}$ and tangent $\overline{MN}$ intersect at point N outside the circle. $PB = 4$ and $PN = 9$. What is the length of $\overline{MN}$? Round to the nearest thousandth.

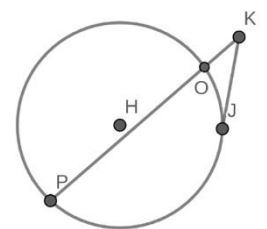


2. Circle A is shown with points M , O and P on the circle. Secant $\overline{PN}$ and tangent $\overline{MN}$ intersect at point N outside the circle. $PO = (2x + 3)$, $ON = 11$ and $MN = 22$. Find the value of x .



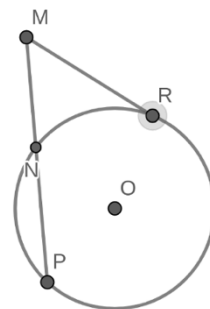
3. What is the length of $\overline{PN}$?

4. Circle H is shown with points J , O and P on the circle. Secant $\overline{KP}$ and tangent $\overline{KJ}$ intersect at point K outside the circle. $PO = 120$, $OK = 8$ and $KJ = x$. Find the value of x .



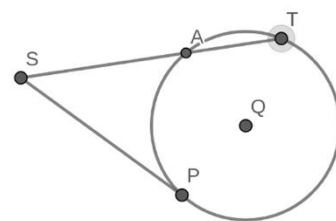
5. Using the same image as question 4, circle H is shown with points J , O , and P on the circle. Secant $\overline{KP}$ and tangent $\overline{KJ}$ intersect at point K outside the circle. $PO = 12 + x$, $OK = 10$ and $KJ = 18$. Find the length of $\overline{KP}$.

6. Circle O is shown with points N , R and P on the circle. Secant $\overline{MP}$ and tangent $\overline{MR}$ intersect at point M outside the circle. $PN = 16$, $MN = 10$ and $MR = 2x$. What is the value of x ? Round to the nearest thousandth.

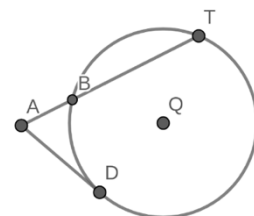


7. Using the same image as question 6, Circle O is shown with points N , R and P on the circle. Secant $\overline{MP}$ and tangent $\overline{MR}$ intersect at point M outside the circle. $PN = x + 17$, $MN = 12$ and $MR = 24$. What is the value of x ?

8. Circle Q is shown with points A , T and P on the circle. Secant $\overline{ST}$ and tangent $\overline{SP}$ intersect at point S outside the circle. $PS = 6x$, $AT = 10$ and $AS = 25$. What is the value of x ? Round to the nearest thousandth.



9. Daniella and Taylor are working on finding the length of $\overline{AD}$. Circle Q is shown with points B , T and D on the circle. Secant $\overline{AT}$ and tangent $\overline{AD}$ intersect at point A outside the circle. $BT = 15$, $AB = 12$ and $AD = 2x$. Their partial work solving for x is shown below. Whose work do you agree with?



Daniella's work	Taylor's work
$(2x)^2 = 15(15 + 12)$ $4x^2 = 15(27)$ $4x^2 = 405$ $x^2 = 101.25$	$(2x)^2 = 12(15 + 12)$ $4x^2 = 12(27)$ $4x^2 = 324$ $x^2 = 81$

10. Using the image in Question 9, what is the length of $\overline{AD}$?

Geometry

Unit 12

Lesson 4 Homework

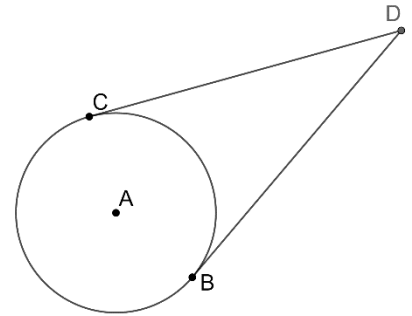
Name _____

Date _____

Period _____

For questions 1 through 4, Circle A is shown. $\overline{CD}$ and $\overline{DB}$ are tangent to the circle.

$$CD = 10x + 30 \text{ and } DB = 4x + 66.$$

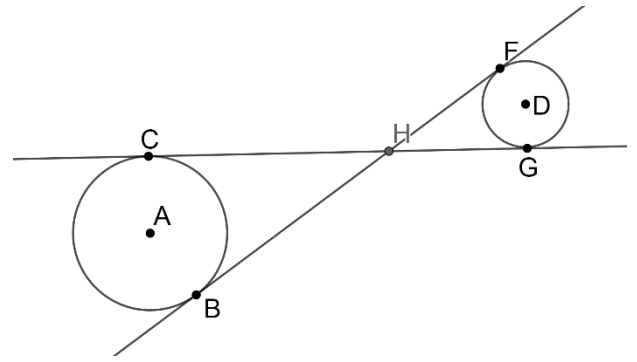


1. What is the value of x ?

2. What is the length of $\overline{CD}$?

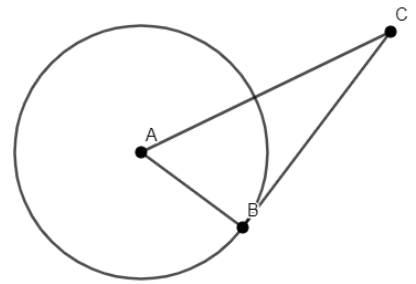
3. What is the length of $\overline{DB}$?

For questions 4 through 10, Circles A and D are shown. Lines $\overline{CG}$ and $\overline{BF}$ intersect at point H . Line $\overline{CG}$ is tangent to circle A at point C and circle D at point G . Line $\overline{BF}$ is tangent to circle A at point B and circle D at point F . $CH = 3x - 8$, $HG = 5y$, $BH = 2x - 3$, $HF = 3y + 2$.



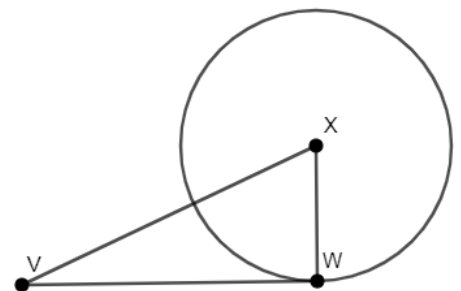
4. Write an equation you can use to solve for x .
5. Solve the equation you wrote in question 4 for x .
6. Write an equation you can use to solve for y .
7. Solve the equation you wrote in question 6 for y .
8. What is the length of $\overline{BF}$?
9. What is the length of $\overline{CH}$?
10. What is the length of $\overline{HF}$?

1. Circle A is shown with point C outside the circle, where point B is on the circle, $AC = 10$, $CB = 8$, and $BA = 6$. Verify if $\overline{BC}$ is tangent to circle A using the Pythagorean Theorem.



Circle X on the right has $\overline{VW}$ as a tangent segment where W is the point of tangency. $VX = 13$ and $VW = 12$.

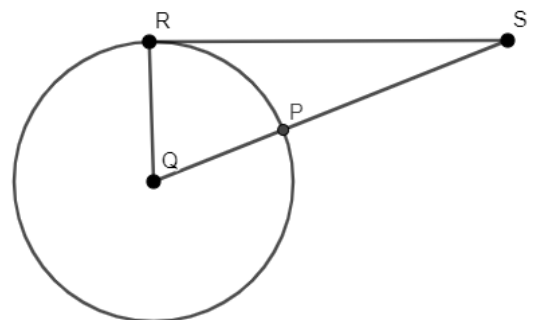
2. What is the length of $\overline{XW}$?



3. What is the area of $\triangle XWV$?

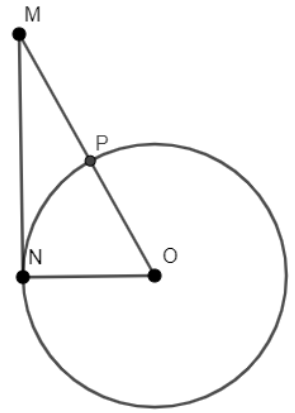
In Circle Q , $\overline{RS}$ is tangent to the circle at point R , $QR = 7$, and $PS = 18$.

4. What is the length of $\overline{RS}$?



5. What is the area of $\triangle QRS$?

In Circle O , $\overline{MN}$ is a tangent segment whose point of tangency is point N . $MN = 15$ and $MP = 6$.

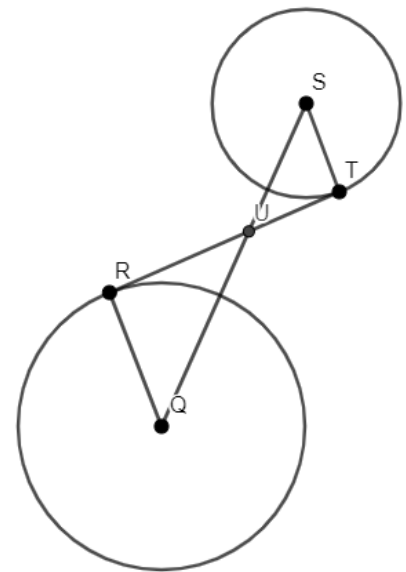


6. What is the length of $\overline{ON}$?

7. What is the length of $\overline{MO}$?

8. What is the perimeter of $\triangle MNO$?

In the figure on the right, $\overline{RT}$ is a common tangent segment of both Circle Q and Circle S . $QR = 15$, $RU = 8$, and $TU = 6$.



9. What is the length of $\overline{ST}$?

10. What is the length of $\overline{QS}$?

Geometry

Unit 12

Lesson 6 Homework

Name _____

Date _____

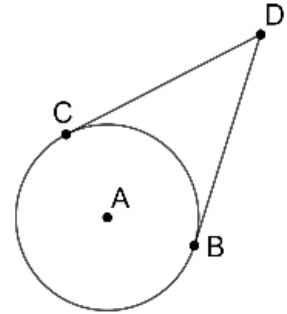
Period _____

For questions 1 through 3, Circle A is shown. $\overline{CD}$ and $\overline{DB}$ are tangent to the circle.
 $CD = 3x + 8$ and $DB = 5x - 16$.

1. What is the value of x ?

2. What is the length of $\overline{CD}$?

3. What is the length of $\overline{DB}$?



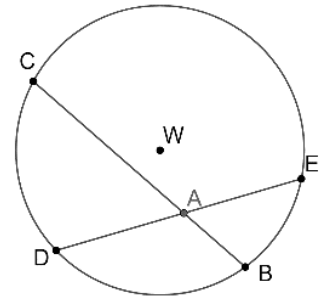
For questions 4 through 7, Circle W is shown. Chords $\overline{CB}$ and $\overline{DE}$ intersect at point A . $CA = 3x + 5$, $AB = 3$, $DA = 4$, and $AE = 2x + 4$.

4. Write an equation you can use to solve for x .

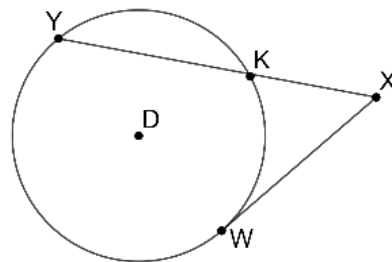
5. Solve the equation you wrote in question 4 for x .

6. What is the length of $\overline{CB}$?

7. What is the length of $\overline{DE}$?



For questions 8 through 10, Circle D is shown. $\overline{XY}$ is a secant to the circle, and intersects tangent $\overline{XW}$ at point X. The length of $\overline{XW} = 6$, $\overline{KX} = 2$, and $\overline{YX} = 3x + 1$.



8. Write an equation that can be used to find the value of x .

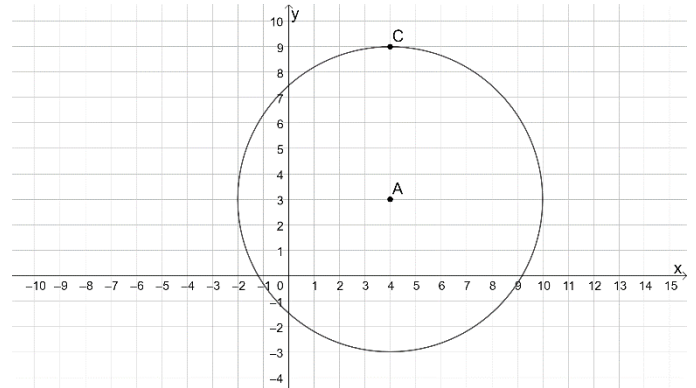
9. Solve the equation you wrote in question 8 to find the value of x .

10. What is the length of $\overline{YX}$?

Geometry
Unit 12
Lesson 7 Homework

Name _____
Date _____
Period _____

For questions 1 and 2, Circle A is shown. Point C lies on the circle.

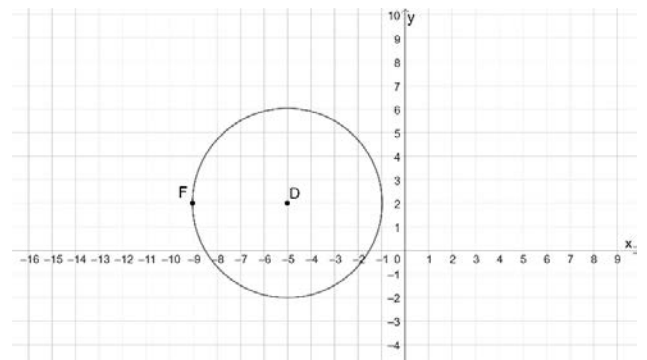


1. Give the key features of the circle.

Center	Diameter

2. Write the equation that represents Circle A .

For questions 3 and 4, Circle D is shown. Point F lies on the circle.



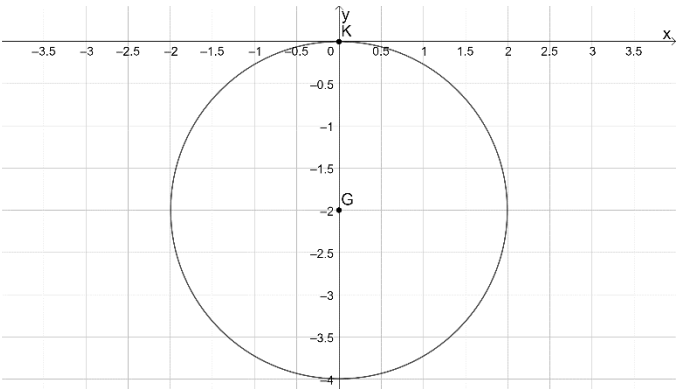
3. Give the key features of the circle.

Center	Diameter

4. Write the equation that represents Circle D .

For questions 5 and 6, Circle G is shown.
Point K lies on the circle.

5. Give the key features of the circle.

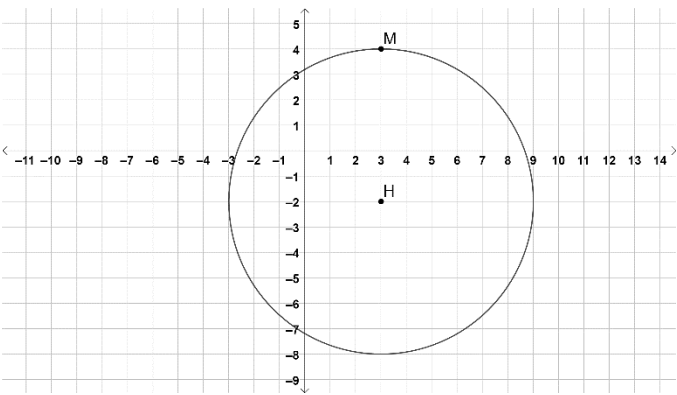


Center	Diameter

6. Write the equation that represents Circle G .

For questions 7 and 8, Circle H is shown.
Point M lies on the circle.

7. Give the key features of the circle.



Center	Diameter

8. Write the equation that represents Circle H .

For questions 1 and 2, consider the equation of the circle represented by $(x + 3)^2 + (y + 4)^2 = 9^2$.

1. Rewrite the equation in general form.

2. Give the key features of the circle.

Center	Diameter	Radius
Domain	Range	

For questions 3 and 4, consider the equation of the circle represented by $(x - 5)^2 + (y + 2)^2 = 5^2$.

3. Rewrite the equation in general form.

4. Give the key features of the circle.

Center	Diameter	Radius
Domain	Range	

For questions 5 and 6, consider the equation of the circle represented by $x^2 + y^2 - 2x - 6y = -6$.

5. Rewrite the equation in center-radius form.

6. Give the key features of the circle.

Center	Diameter	Radius
Domain	Range	

For questions 7 and 8, consider the equation of the circle represented by $x^2 + y^2 - 20y = 21$.

7. Rewrite the equation in center-radius form.

8. Give the key features of the circle.

Center	Diameter	Radius
Domain	Range	

For questions 9 and 10, consider the equation of the circle represented by $x^2 + y^2 - 10x = -24$.

9. Rewrite the equation in center-radius form.

10. Give the key features of the circle.

Center	Diameter	Radius
Domain	Range	

Geometry
Unit 12
Lesson 9 Homework

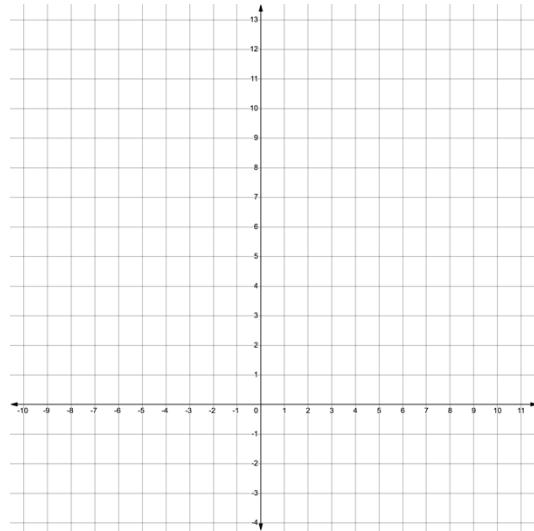
Name _____
Date _____
Period _____

For questions 1 and 2, consider that the equation of circle S is $(x - 2)^2 + (y - 7)^2 = 25$.

1. Give the key features of the circle.

Center	Radius
Domain	Range

2. Graph circle S .

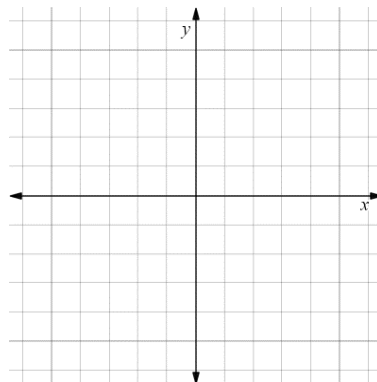


For questions 3 and 4, consider that the equation of circle K is $(x + 1)^2 + (y - 2)^2 = 4$.

3. Give the key features of the circle.

Center	Radius
Domain	Range

4. Graph circle K .

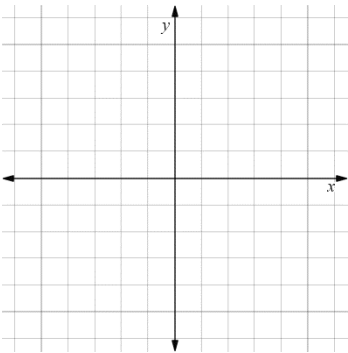


For questions 5 through 7, consider that the equation of circle L is $x^2 + 2x + y^2 + 2y = 7$.

5. Rewrite the equation of the circle in center-radius form.
6. Give the key features of the circle.

Center	Radius
Domain	Range

7. Graph circle L .



For questions 8 through 10, consider that the equation of circle Z is $x^2 - 6x + y^2 + 4y - 12 = 0$.

8. Rewrite the equation of the circle in center-radius form.
9. Give the key features of the circle.

Center	Radius
Domain	Range

10. Graph circle Z .

